Custom E-Commerce Project — Export Summary

Project Overview

Custom e-commerce platform for a small Brazilian business that sells flavorings.

Main Goals

- Modern custom storefront
- PIX payments
- Shipping integration
- NFe emission
- CMS/admin panel for owner
- Mobile-first experience
- Production deployment in November

Technology Stack

Frontend

- Bun
- TypeScript
- React
- Next.js
- TailwindCSS
- shadcn/ui
- Zustand
- TanStack Query

Backend

- Python
- FastAPI
- SQLModel / SQLAlchemy
- Pydantic
- Alembic

Database & Storage

- Supabase PostgreSQL
- Supabase Storage
- Supabase Auth

Infrastructure

- Vercel (frontend)

- Railway or Render (backend)

Payments

- Sicredi PIX API

Shipping

- Melhor Envio

Fiscal / NFe

- eNotas or Nuvem Fiscal

Project Scope

Customer Features

- Product catalog
- Categories
- Product pages
- Search
- Cart
- Checkout
- PIX payment
- Shipping calculation
- Free shipping over R$ 300,00
- Responsive/mobile-first UI

Admin/CMS Features

- Product CRUD
- Image uploads
- Stock management
- Order management
- Shipping management
- Invoice management
- Dashboard

Fiscal Features

- NFe emission
- DANFE/XML generation and storage
- CPF/CNPJ validation

Shipping Features

- Shipping quotes

- Delivery options
- Tracking codes
- Free shipping rules

Recommended Architecture

```
Next.js Frontend + Admin
        ↓
FastAPI Backend
        ↓
Supabase PostgreSQL
        ↓
Supabase Storage
        ↓
Sicredi PIX
        ↓
Melhor Envio
        ↓
eNotas / Nuvem Fiscal
```

Pricing Table

Plan	Price	Includes
Essential	R$ 12.000	Complete e-commerce with catalog, cart, PIX checkout, CMS/admin, shipping calculation, free shipping rules, NFe integration, responsive design and production deployment.
Professional	R$ 18.000	Everything from Essential + advanced admin dashboard, order management, stock control, tracking, UX/UI improvements, SEO basics and operational automations.
Premium	R$ 25.000+	Everything from Professional + custom features, advanced dashboards, extra integrations, performance optimization, priority support and future scalability.

Monthly Maintenance

Service	Price	Includes
Basic Maintenance	R$ 500/month	Hosting, monitoring, backups and minor fixes.

Service	Price	Includes
Full Support	R$ 1.000–2.000/ month	Continuous support, integrations maintenance, improvements and operational monitoring.

Timeline

```
Development: May → October
Production Deployment & Stabilization: November
```

Sprint Roadmap (6 Months)

MAY — Foundation

Sprint 1 — Infrastructure & Setup

Deliverables

- Next.js setup
- FastAPI setup
- Supabase setup
- Environments
- Docker
- CI/CD
- Repository structure
- Authentication architecture

Sprint 2 — Database & UI Foundation

Deliverables

- Database schema
- Migrations
- Design system
- Tailwind
- shadcn/ui
- Admin shell
- API structure
- RBAC basics

JUNE — Commerce Core

Sprint 3 — Product System

Deliverables

- Products API
- Categories
- Product detail pages
- Image upload
- Storage integration

Sprint 4 — Cart & CMS

Deliverables

- Cart system
- Zustand store
- Cart persistence
- Admin CRUD
- Product editing
- Stock management

JULY — Checkout & Payments

Sprint 5 — Checkout System

Deliverables

- Checkout UI
- Address forms
- Order creation
- Validation
- Totals calculation

Sprint 6 — PIX Integration

Deliverables

- Sicredi integration
- QR code flow
- Payment polling/webhooks
- Payment table
- Order status updates

AUGUST — Logistics & Fiscal

Sprint 7 — Shipping

Deliverables

- Melhor Envio integration
- Shipping calculations
- Freight options
- Free shipping rules
- Tracking structure

Sprint 8 — NFe

Deliverables

- eNotas/Nuvem Fiscal integration
- CPF/CNPJ validation
- Invoice generation
- XML/PDF storage
- Fiscal workflows

SEPTEMBER — Operations

Sprint 9 — Order Management

Deliverables

- Admin order dashboard
- Filtering
- Status flows
- Shipping updates
- Invoice visualization

Sprint 10 — Operational Hardening

Deliverables

- Audit logs
- Retry systems
- Webhook idempotency
- Permission validation
- Security review

- Backups

OCTOBER — Production Readiness

Sprint 11 — QA & Optimization

Deliverables

- Responsive fixes
- Performance optimization
- SEO
- Accessibility
- Bug fixing
- UX polish

Sprint 12 — Launch Preparation

Deliverables

- Production environment
- Monitoring
- Observability
- Analytics
- Final testing
- Deployment rehearsal
- Documentation

NOVEMBER — Production Deployment

Deliverables

- Production release
- Domain/DNS setup
- SSL
- Monitoring
- Support
- Hotfixes
- Operational follow-up

Kanban Structure

Recommended Columns

Column	Purpose
Backlog	Future ideas/tasks
Ready	Refined and approved tasks
In Progress	Current development
Review	QA/code review/testing
Blocked	Waiting dependency
Done	Finished and validated

Main EPICS

1. Foundation & Infrastructure
2. Storefront
3. CMS / Admin
4. Cart & Checkout
5. Payments (PIX)
6. Shipping
7. NFe / Fiscal
8. Orders & Operations
9. Security & Reliability
10. QA & Optimization
11. Deployment & Production

EPIC 1 — Foundation & Infrastructure

Goal

Establish the technical foundation:

- repositories
- architecture
- auth
- database
- CI/CD
- environments
- base frontend/backend structure

TASK 1 — Project Architecture & Repository Setup

Subtasks

- Create monorepo structure
- Configure Bun workspace
- Initialize Next.js frontend
- Initialize FastAPI backend
- Configure environment variables
- Configure TypeScript
- Configure Python environment
- Configure linting/formatting
- Setup path aliases
- Create README

TASK 2 — Infrastructure & Deployment Base

Subtasks

- Create Vercel project
- Create Railway/Render backend project
- Configure Supabase project
- Configure PostgreSQL connection
- Configure Supabase Storage
- Setup Docker files
- Setup Docker Compose
- Configure environments
- Configure domain planning
- Configure SSL strategy

TASK 3 — CI/CD Pipeline

Subtasks

- Configure GitHub repository
- Configure branch strategy
- Setup GitHub Actions
- Configure frontend build pipeline
- Configure backend build pipeline
- Configure lint checks
- Configure deployment automation
- Configure environment secrets

TASK 4 — Database Foundation

Subtasks

- Design ERD
- Create products table
- Create categories table
- Create users table
- Create orders table
- Create order_items table
- Create payments table
- Create invoices table
- Configure foreign keys
- Configure indexes
- Create migrations

TASK 5 — Backend Core Structure

Subtasks

- Setup FastAPI structure
- Configure routing architecture
- Configure dependency injection
- Setup SQLModel/SQLAlchemy
- Configure async sessions
- Create repository pattern
- Create service layer
- Configure Pydantic schemas
- Configure logging

TASK 6 — Authentication & Authorization

Subtasks

- Configure Supabase Auth
- Configure JWT validation
- Create login flow
- Configure protected routes
- Create admin role system
- Configure RBAC middleware
- Configure backend auth guards

TASK 7 — Frontend Base Architecture

Subtasks

- Configure TailwindCSS
- Configure shadcn/ui
- Configure responsive layout
- Setup Zustand
- Setup TanStack Query
- Configure API client
- Configure loading states
- Configure error boundaries

TASK 8 — Admin Base Layout

Subtasks

- Create admin layout
- Create sidebar navigation
- Create dashboard shell
- Configure protected admin routes
- Create reusable table components
- Create reusable form components

TASK 9 — Security & Environment Setup

Subtasks

- Configure secret management
- Configure API rate limiting
- Configure secure headers
- Configure input sanitization
- Configure backup strategy
- Configure monitoring preparation

TASK 10 — Documentation & Technical Standards

Subtasks

- Document architecture decisions
- Define naming conventions
- Define Git workflow
- Create onboarding guide
- Create deployment checklist

Estimated Project Complexity

```
Estimated workload: 400–900 development hours
```

Strategic Notes

Recommended Priorities

```
Commerce Core
→ Payments
→ Shipping
→ Fiscal
→ Polish
```

Recommended Architecture Style

```
Modular Monolith
NOT microservices
```

Biggest Project Risk

```
Scope creep
```

Lock early:

- payment provider
- shipping provider
- fiscal provider
- admin requirements

Recommended Final Stack

Frontend

- Next.js
- TypeScript
- Bun
- Zustand
- TanStack Query

- shadcn/ui
- TailwindCSS

Backend

- FastAPI
- SQLModel
- Alembic
- Pydantic
- httpx

Infra

- Supabase
- Vercel
- Railway/Render

Integrations

- Sicredi PIX
- Melhor Envio
- eNotas / Nuvem Fiscal